

Отчет по paper-экспериментам для статьи об emotionally bounded-rational agents

Источник: paper_experiment_bundle_20260511_224550.zip. Подготовлено на основе CSV, config.json, run_log.txt и plots из переданного архива.

1. Статус пакета и валидации

Пакет содержит полный набор обязательных paper-прогонов для текущей статьи: S1, S2, S3, S4. В дополнение проведен optional S3 по всем играм. Финальный validation log содержит VALIDATION OK. Ниже приведен инвентарь запусков.

run_dir	scenario	profile	games	episodes	rounds	conditions	summary_rows	plots_count	ok
results/ 02_paper/ 20260511_2 01116_paper _S1	S1	paper	Prisoners Dilemma; Battle of Sexes; Ultimatum Game	20	200	48	48	48	1
results/ 02_paper/ 20260511_2 04523_paper _S2	S2	paper	Prisoners Dilemma; Battle of Sexes; Ultimatum Game	20	200	27	27	152	1
results/ 02_paper/ 20260511_2 10148_paper _S3	S3	paper	Prisoners Dilemma	20	200	27	27	18	1
results/ 02_paper_op tional/ 20260511_2 13155_paper _S3	S3	paper	Prisoners Dilemma; Battle of Sexes; Ultimatum Game	20	200	81	81	48	1
results/ 02_paper/ 20260511_2 11733_paper _S4	S4	paper	Prisoners Dilemma; Battle of Sexes; Ultimatum Game	20	200	12	12	168	1
results/ 00_smoke/ 20260511_1 93635_quick _S0	S0	quick	Prisoners Dilemma	1	20	1	1	13	1
results/ 01_standard/ 20260511_1 94003_stand ard_S1	S1	standard	Prisoners Dilemma; Battle of Sexes; Ultimatum Game	10	100	48	48	24	1
results/ 01_standard/ 20260511_1 95447_stand ard_S2	S2	standard	Prisoners Dilemma; Battle of Sexes; Ultimatum Game	10	100	27	27	76	1
results/ 01_standard/ 20260511_2 00159_stand ard_S3	S3	standard	Prisoners Dilemma	10	100	27	27	9	1
results/ 01_standard/ 20260511_2 00637_stand ard_S4	S4	standard	Prisoners Dilemma; Battle of Sexes; Ultimatum Game	10	100	12	12	84	1

2. Краткое содержание экспериментов

- S1: полная ordered cross-type matrix по Prisoners Dilemma, Battle of Sexes и Ultimatum Game.
- S2: адаптация к fixed strategies через сравнение первой и последней трети эпизода.
- S3: personality x emotional intensity sweep; core run для Prisoners Dilemma и optional run для всех игр.
- S4: internal-state dynamics/collapse diagnostics: round-level trajectories, cumulative payoff и event values.
- Все paper-прогоны выполнены с profile=paper: 20 episodes, 200 rounds; standard-прогоны сохранены для технической проверки.

3. S1: cross-type matchup matrix

S1 содержит 48 условий: 3 игры x 16 ordered pairs. Главный смысл для статьи — не заявлять универсальное превосходство гибридного агента, а показать, где payoff, game-specific metrics и internal state расходятся.

Топ-5 условий по pair payoff для каждой игры:

game_name	condition_label	pair_total_payoff_mean	agent_1_payoff_mean	agent_2_payoff_mean	game_metric_name	game_metric_value
Battle of Sexes	fixed-tit_for_tat_vs_fixed-tit_for_tat	1000	600	400	agreement rate	0.0
Battle of Sexes	rational_vs_fixed-tit_for_tat	683.250	273.850	409.400	agreement rate	0.0
Battle of Sexes	fixed-tit_for_tat_vs_rational	678.750	272.100	406.650	agreement rate	0.0
Battle of Sexes	rational_vs_rational	550.750	265.200	285.550	agreement rate	0.55075
Battle of Sexes	fixed-tit_for_tat_vs_emotional-neutral-neutral	529	212.150	316.850	agreement rate	0.0
Prisoners Dilemma	fixed-tit_for_tat_vs_fixed-tit_for_tat	1200	600	600	mutual cooperation rate	1.0
Prisoners Dilemma	emotional-neutral-neutral_vs_fixed-tit_for_tat	901.350	452.300	449.050	mutual cooperation rate	0.2815000000000003
Prisoners Dilemma	hybrid-neutral-neutral_vs_emotional-neutral-neutral	897.900	453.200	444.700	mutual cooperation rate	0.24175
Prisoners Dilemma	hybrid-neutral-neutral_vs_hybrid-neutral-neutral	896.550	441.150	455.400	mutual cooperation rate	0.24925
Prisoners Dilemma	emotional-neutral-neutral_vs_hybrid-neutral-neutral	896.450	458.100	438.350	mutual cooperation rate	0.23675
Ultimatum Game	hybrid-neutral-neutral_vs_rational	1183	572.058	610.942	acceptance rate / mean offer	0.5915 / 5.433613172814509
Ultimatum Game	rational_vs_rational	1162.500	591.522	570.977	acceptance rate / mean offer	0.58125 / 5.08537772851627
Ultimatum Game	fixed-tit_for_tat_vs_rational	1156.500	578.250	578.250	acceptance rate / mean offer	0.57825 / 0.0
Ultimatum Game	emotional-neutral-neutral_vs_rational	1144	550.488	593.512	acceptance rate / mean offer	0.5720000000000001 / 5.638919480345601
Ultimatum Game	emotional-neutral-neutral_vs_hybrid-neutral-neutral	703.500	343.343	360.158	acceptance rate / mean offer	0.35175 / 5.412945686870658

Выбранные сравнения для текста статьи:

game_name	condition_label	pair_total_payoff_mean	agent_1_payoff_mean	agent_2_payoff_mean	agent_1_mean_overall_state	agent_2_mean_overall_state
Prisoners Dilemma	hybrid-neutral-neutral_vs_hybrid-neutral-neutral	896.550	441.150	455.400	-1.013	-0.994
Prisoners Dilemma	hybrid-neutral-neutral_vs_rational	832.750	313.500	519.250	-1.008	-0.227
Prisoners Dilemma	rational_vs_rational	734.250	372.750	361.500	-0.111	-0.092
Prisoners Dilemma	emotional-neutral-neutral_vs_rational	830.350	327.800	502.550	-0.809	-0.190

Prisoners Dilemma	fixed-tit_for_tat_vs_fixed-tit_for_tat	1200	600	600	-0.750	-0.694
Battle of Sexes	hybrid-neutral-neutral_vs_hybrid-neutral-neutral	492	245.150	246.850	-1.103	-1.058
Battle of Sexes	hybrid-neutral-neutral_vs_rational	477.250	226.050	251.200	-1.072	-0.388
Battle of Sexes	rational_vs_rational	550.750	265.200	285.550	-0.409	-0.390
Battle of Sexes	emotional-neutral-neutral_vs_rational	509	243.550	265.450	-0.798	-0.375
Battle of Sexes	fixed-tit_for_tat_vs_fixed-tit_for_tat	1000	600	400	-0.850	-0.792
Ultimatum Game	hybrid-neutral-neutral_vs_hybrid-neutral-neutral	697.500	341.750	355.750	-1.048	-1.022
Ultimatum Game	hybrid-neutral-neutral_vs_rational	1183	572.058	610.942	-1.071	-0.176
Ultimatum Game	rational_vs_rational	1162.500	591.522	570.977	-0.170	-0.167
Ultimatum Game	emotional-neutral-neutral_vs_rational	1144	550.488	593.512	-0.800	-0.193
Ultimatum Game	fixed-tit_for_tat_vs_fixed-tit_for_tat	0	0	0	-0.708	-0.722

- В Prisoners Dilemma максимальный парный выигрыш дает fixed tit-for-tat vs fixed tit-for-tat (1200; mutual cooperation=1.0). Это полезный sanity baseline.
- В Prisoners Dilemma hybrid vs rational дает высокий pair payoff (832.75), но payoff распределен асимметрично: rational получает 519.25, hybrid 313.50. Это хороший пример эксплуатации/асимметрии, а не победы гибрида.
- В Ultimatum Game наиболее информативны условия с rational opponent/responder: hybrid vs rational (1183.0) и rational vs rational (1162.5).
- Для Battle of Sexes fixed-strategy условия дают высокие payoff, но agreement_rate часто 0. Поэтому эти строки лучше использовать как технический baseline, а не как центральное доказательство координации.

4. S2: adaptation to fixed strategies

S2 позволяет аккуратно поддержать H1 в виде проху-adaptation evidence: сравнение метрики и payoff первой и последней трети эпизода. Ниже агрегаты по 20 эпизодам.

game_name	agent_1_type	fixed_strategy_2	metric_name	first_metric	last_metric	delta_metric_mean	first_payoff	last_payoff	delta_payoff_mean
Battle of Sexes	emotional	always_cooperate	agreement_flag	0	0	0	96.450	97.950	1.500
Battle of Sexes	emotional	always_defect	agreement_flag	0	0	0	67.400	58.700	-8.700
Battle of Sexes	emotional	tit_for_tat	agreement_flag	0	0	0	68.350	63.800	-4.550
Battle of Sexes	emotional_rational	always_cooperate	agreement_flag	0	0	0	99.300	103.200	3.900
Battle of Sexes	emotional_rational	always_defect	agreement_flag	0	0	0	65.700	67.900	2.200
Battle of Sexes	emotional_rational	tit_for_tat	agreement_flag	0	0	0	67.250	66.500	-0.750
Battle of Sexes	rational	always_cooperate	agreement_flag	0	0	0	110.100	151.350	41.250
Battle of Sexes	rational	always_defect	agreement_flag	0	0	0	75.700	104.800	29.100
Battle of Sexes	rational	tit_for_tat	agreement_flag	0	0	0	76.450	104.800	28.350
Prisoners Dilemma	emotional	always_cooperate	cooperation_flag_1	0.488	0.521	0.033	265.600	261.200	-4.400
Prisoners Dilemma	emotional	always_defect	cooperation_flag_1	0.489	0.530	0.041	33.700	31	-2.700
Prisoners Dilemma	emotional	tit_for_tat	cooperation_flag_1	0.486	0.530	0.043	148.350	150.100	1.750
Prisoners Dilemma	emotional_rational	always_cooperate	cooperation_flag_1	0.505	0.445	-0.059	263.400	271.200	7.800
Prisoners Dilemma	emotional_rational	always_defect	cooperation_flag_1	0.499	0.458	-0.042	33.050	35.800	2.750
Prisoners Dilemma	emotional_rational	tit_for_tat	cooperation_flag_1	0.503	0.471	-0.032	149.900	142.050	-7.850
Prisoners Dilemma	rational	always_cooperate	cooperation_flag_1	0.452	0.206	-0.245	270.400	302.800	32.400
Prisoners Dilemma	rational	always_defect	cooperation_flag_1	0.427	0.206	-0.221	37.800	52.400	14.600
Prisoners Dilemma	rational	tit_for_tat	cooperation_flag_1	0.433	0.211	-0.223	142.050	105.700	-36.350

Ultimatum Game	emotional	always_cooperate	fairness_proxy	0.853	0.849	-0.005	0	0	0
Ultimatum Game	emotional	always_defect	fairness_proxy	0.853	0.849	-0.005	0	0	0
Ultimatum Game	emotional	tit_for_tat	fairness_proxy	0.853	0.849	-0.005	0	0	0
Ultimatum Game	emotional_rational	always_cooperate	fairness_proxy	0.847	0.851	0.004	0	0	0
Ultimatum Game	emotional_rational	always_defect	fairness_proxy	0.847	0.851	0.004	0	0	0
Ultimatum Game	emotional_rational	tit_for_tat	fairness_proxy	0.847	0.851	0.004	0	0	0
Ultimatum Game	rational	always_cooperate	fairness_proxy	0.849	0.842	-0.006	0	0	0
Ultimatum Game	rational	always_defect	fairness_proxy	0.849	0.842	-0.006	0	0	0
Ultimatum Game	rational	tit_for_tat	fairness_proxy	0.849	0.842	-0.006	0	0	0

- В Prisoners Dilemma rational agent резко снижает cooperation против fixed opponents и увеличивает payoff против always_cooperate и always_defect; это payoff-oriented adaptation.
- Hybrid agent против always_cooperate в PD снижает cooperation умеренно и увеличивает payoff; emotional agent, наоборот, немного повышает cooperation и теряет payoff.
- Для Battle of Sexes agreement_flag остается 0 в fixed-strategy матчах, поэтому вывод по S2/BoS лучше формулировать через payoff adaptation, а не через agreement.
- В Ultimatum Game условия с fixed_strategy дают нулевой payoff и не должны использоваться как substantive evidence адаптации; это техническое ограничение текущей постановки fixed baseline.

5. S3: personality x emotional intensity

Core S3 проведен для Prisoners Dilemma; optional S3 — для всех игр. Центральный вывод: эффекты personality/intensity есть, но они умеренные; в статье лучше писать о sensitivity analysis, а не о сильной универсальной закономерности.

Core S3 по Prisoners Dilemma, среднее по fixed opponents:

game_name	personality	intensity	pair_payoff_mean	agent_1_payoff_mean	agent_1_state_mean	negative_state_share
Prisoners Dilemma	pessimistic	neutral	905.250	448.583	-1.066	1.000
Prisoners Dilemma	neutral	low	903.633	445.733	-1.118	0.999
Prisoners Dilemma	pessimistic	low	902.817	448.700	-1.127	1
Prisoners Dilemma	optimistic	neutral	899.567	447.200	-0.973	0.996
Prisoners Dilemma	optimistic	low	899.117	448.350	-1.054	0.999
Prisoners Dilemma	optimistic	high	898.250	445.750	-0.970	0.995
Prisoners Dilemma	neutral	high	896.383	451.067	-1.019	0.998
Prisoners Dilemma	pessimistic	high	895.383	453.233	-1.024	0.999
Prisoners Dilemma	neutral	neutral	891.950	450.683	-1.052	0.999

Optional S3 по всем играм, среднее по fixed opponents:

game_name	personality	intensity	pair_payoff_mean	agent_1_payoff_mean	agent_1_state_mean	negative_state_share
Battle of Sexes	neutral	neutral	504.750	236.600	-1.104	0.999
Battle of Sexes	optimistic	low	502.917	235.167	-1.111	0.999
Battle of Sexes	neutral	high	500.750	234.917	-1.088	0.999
Battle of Sexes	pessimistic	neutral	500.417	234	-1.140	1
Battle of Sexes	neutral	low	498.500	232.733	-1.138	0.999
Battle of Sexes	pessimistic	high	498.500	232.483	-1.107	1
Battle of Sexes	optimistic	neutral	497.167	233.133	-1.082	0.999
Battle of Sexes	pessimistic	low	497	232.500	-1.179	1
Battle of Sexes	optimistic	high	496.583	232.833	-1.088	0.999
Prisoners Dilemma	pessimistic	neutral	905.250	448.583	-1.066	1.000
Prisoners Dilemma	neutral	low	903.633	445.733	-1.118	0.999
Prisoners Dilemma	pessimistic	low	902.817	448.700	-1.127	1
Prisoners Dilemma	optimistic	neutral	899.567	447.200	-0.973	0.996
Prisoners Dilemma	optimistic	low	899.117	448.350	-1.054	0.999
Prisoners Dilemma	optimistic	high	898.250	445.750	-0.970	0.995
Prisoners Dilemma	neutral	high	896.383	451.067	-1.019	0.998
Prisoners Dilemma	pessimistic	high	895.383	453.233	-1.024	0.999
Prisoners Dilemma	neutral	neutral	891.950	450.683	-1.052	0.999
Ultimatum Game	neutral	high	0	0	-0.989	0.998
Ultimatum Game	neutral	low	0	0	-1.009	0.999
Ultimatum Game	neutral	neutral	0	0	-0.999	0.999
Ultimatum Game	optimistic	high	0	0	-1.021	0.999
Ultimatum Game	optimistic	low	0	0	-1.027	0.999
Ultimatum Game	optimistic	neutral	0	0	-1.013	0.999
Ultimatum Game	pessimistic	high	0	0	-1.027	1
Ultimatum Game	pessimistic	low	0	0	-1.031	1
Ultimatum Game	pessimistic	neutral	0	0	-1.006	0.998

- В PD максимальный средний pair payoff среди personality/intensity cells дает pessimistic-neutral, но лучшее internal state среди топовых конфигураций ближе к optimistic-high/optimistic-neutral.

- Для BoS различия между personality/intensity cells невелики; это стоит описывать как слабый или умеренный эффект.
- Для UG optional S3 с fixed opponents дает нулевой pair payoff; использовать эти строки как limitation/technical artifact.

6. S4: internal-state dynamics / collapse diagnostics

S4 дает материал для Discussion: даже при высоком payoff internal state может оставаться отрицательным, а negative_state_share — близким к 1. Поэтому для статьи лучше подчеркивать относительные различия и trade-off, а не абсолютную психологическую валидность шкалы.

game_name	condition_label	pair_total_payoff_mean	agent_1_payoff_mean	agent_2_payoff_mean	agent_1_mean_overall_state	agent_2_mean_overall_state	agent_1_negative_state_share	game_metric_name	game_metric_value
Prisoners Dilemma	hybrid-neutral-neutral_vs_rational	832.750	313.500	519.250	-1.008	-0.227	0.998	mutual cooperation rate	0.15675
Prisoners Dilemma	hybrid-neutral-neutral_vs_fixed-always_defect	695.350	101.550	593.800	-1.054	-0.624	0.999	mutual cooperation rate	0.0
Prisoners Dilemma	hybrid-neutral-neutral_vs_fixed-tit_for_tat	884.100	443.300	440.800	-1.062	-0.640	0.999	mutual cooperation rate	0.2575
Prisoners Dilemma	emotional-neutral-neutral_vs_rational	830.350	327.800	502.550	-0.809	-0.190	0.999	mutual cooperation rate	0.159
Battle of Sexes	hybrid-neutral-neutral_vs_rational	477.250	226.050	251.200	-1.072	-0.388	0.998	agreement rate	0.47725
Battle of Sexes	hybrid-neutral-neutral_vs_fixed-always_defect	497.750	199.100	298.650	-1.101	-0.726	0.999	agreement rate	0.0
Battle of Sexes	hybrid-neutral-neutral_vs_fixed-tit_for_tat	499.250	200.350	298.900	-1.103	-0.723	0.999	agreement rate	0.0
Battle of Sexes	emotional-neutral-neutral_vs_rational	509	243.550	265.450	-0.798	-0.375	0.999	agreement rate	0.509
Ultimatum Game	hybrid-neutral-neutral_vs_rational	1183	572.058	610.942	-1.071	-0.176	1.000	acceptance rate / mean offer	0.5915 / 5.433613172814509
Ultimatum Game	hybrid-neutral-neutral_vs_fixed-always_defect	0	0	0	-0.999	-0.752	0.999	acceptance rate / mean offer	0.0 / 5.422996938023452
Ultimatum Game	hybrid-neutral-neutral_vs_fixed-tit_for_tat	0	0	0	-0.999	-0.752	0.999	acceptance rate / mean offer	0.0 / 5.422996938023452
Ultimatum Game	emotional-neutral-neutral_vs_rational	1144	550.488	593.512	-0.800	-0.193	0.999	acceptance rate / mean offer	0.5720000000000001 / 5.638919480345601

- В PD hybrid vs tit-for-tat дает более высокий pair payoff, чем hybrid vs always_defect; это ожидаемо и полезно для иллюстрации state dynamics.
- В PD hybrid vs rational и emotional vs rational дают близкий pair payoff, но emotional agent имеет менее отрицательный mean overall state. Это честное ограничение текущего гибридного механизма.
- В UG hybrid/emotional vs rational дают высокие payoffs, но fixed-strategy variants дают нулевой payoff; это снова нужно отметить как ограничение fixed baseline для UG.

7. Ограничения и guardrails для статьи

- Не утверждать, что эти прогоны доказывают H2 или human-like behavior. Они дают поведенческие трейсы и внутренние диагностические переменные для будущей проверки.
- Не писать, что hybrid agent универсально превосходит rational/emotional baselines. Результаты зависят от игры, оппонента и выбранной метрики.
- Внутреннее состояние в текущем коде часто отрицательное; это не внешний психологический тест, а внутренняя диагностическая шкала модели.
- UG fixed-strategy conditions в S2/S3 дают нулевой payoff и не подходят как центральное доказательство адаптации в Ultimatum Game.

- Для статьи предпочтительно использовать S1/S4 по UG с rational opponent и общие графики offer/acceptance, а не fixed-strategy UG rows.

8. Рекомендуемые таблицы и графики для статьи

Таблицы, подготовленные в папке tables/:

- run_inventory.csv — проверка состава прогонов.
- S1_top_pair_payoff_by_game.csv и S1_selected_comparisons.csv — основной материал для Results по matchup matrix.
- S2_adaptation_first_vs_last_third.csv — материал для proxy-adaptation/H1.
- S3_core_PD_personality_intensity_summary.csv и S3_optional_all_games_personality_intensity_summary.csv — материал для H3.
- S4_internal_state_diagnostics_summary.csv — материал для Discussion/internal-state dynamics.
- interpretation_guardrails.csv — список ограничений, которые важно явно учесть при написании статьи.

Отобранные рисунки скопированы в selected_figures/. Их источники указаны в selected_figure_manifest.csv.

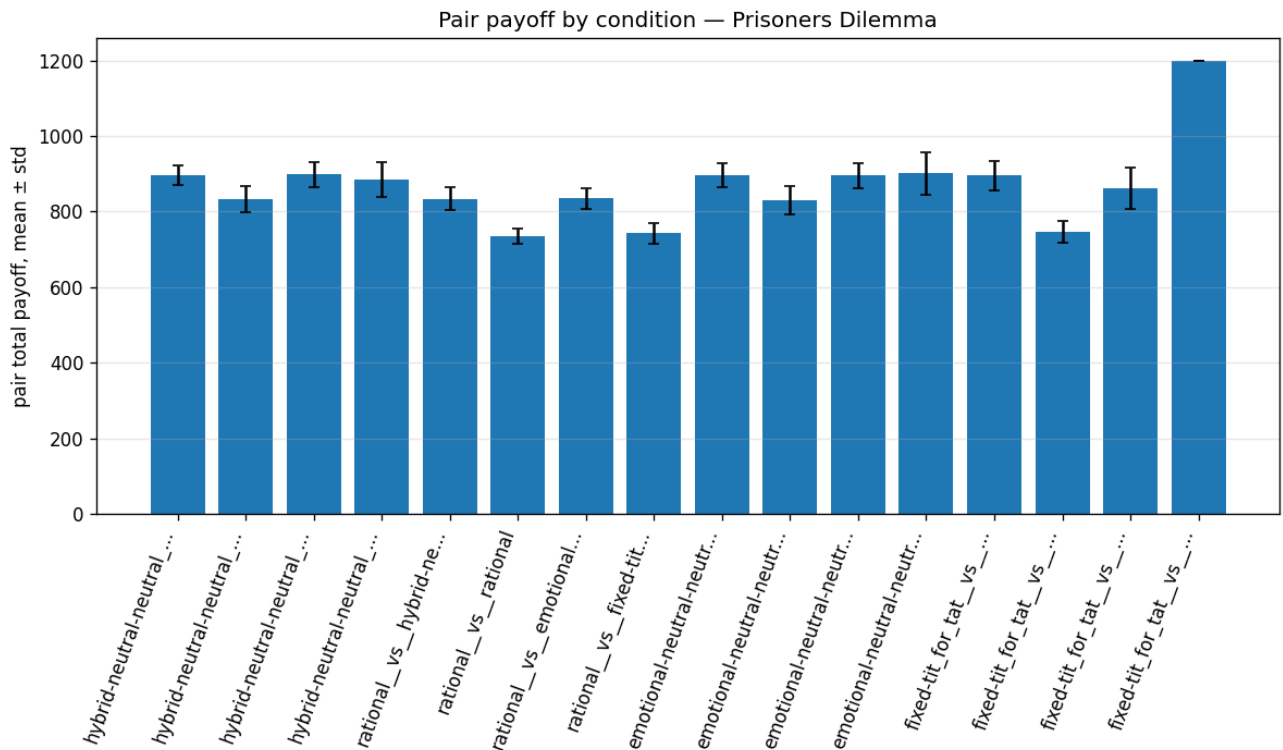
figure_file	source_run	purpose
payoff_bar_Prisoners_Dilemma.png	results/02_paper/20260511_201116_paper_S1	S1 matchup matrix: payoff/internal state/game-specific metric
payoff_bar_Battle_of_Sexes.png	results/02_paper/20260511_201116_paper_S1	S1 matchup matrix: payoff/internal state/game-specific metric
payoff_bar_Ultimatum_Game.png	results/02_paper/20260511_201116_paper_S1	S1 matchup matrix: payoff/internal state/game-specific metric
overall_state_bar_Prisoners_Dilemma.png	results/02_paper/20260511_201116_paper_S1	S1 matchup matrix: payoff/internal state/game-specific metric
game_metric_Prisoners_Dilemma_pd_mutual_cooperation_rate.png	results/02_paper/20260511_201116_paper_S1	S1 matchup matrix: payoff/internal state/game-specific metric
game_metric_Ultimatum_Game_ug_acceptance_rate.png	results/02_paper/20260511_201116_paper_S1	S1 matchup matrix: payoff/internal state/game-specific metric
adaptation_first_vs_last.png	results/02_paper/20260511_204523_paper_S2	S2 adaptation to fixed strategies
20260511_204523_paper_S2_payoff_bar_Prisoners_Dilemma.png	results/02_paper/20260511_204523_paper_S2	S2 adaptation to fixed strategies
20260511_204523_paper_S2_game_metric_Prisoners_Dilemma_pd_mutual_cooperation_rate.png	results/02_paper/20260511_204523_paper_S2	S2 adaptation to fixed strategies
S3_heatmap_pair_payoff_Prisoners_Dilemma.png	results/02_paper/20260511_210148_paper_S3	S3 personality x intensity core PD
boxplot_Prisoners_Dilemma_pair_total_payoff.png	results/02_paper/20260511_210148_paper_S3	S3 personality x intensity core PD
20260511_210148_paper_S3_overall_state_bar_Prisoners_Dilemma.png	results/02_paper/20260511_210148_paper_S3	S3 personality x intensity core PD
S3_heatmap_pair_payoff_Battle_of_Sexes.png	results/02_paper_optional/20260511_213155_paper_S3	Optional S3 all-games extension
S3_heatmap_pair_payoff_Ultimatum_Game.png	results/02_paper_optional/20260511_213155_paper_S3	Optional S3 all-games extension
timeseries_Prisoners_Dilemma_hybrid-neutral-neutral_vs_fixed-always_defect_agent1.png	results/02_paper/20260511_211733_paper_S4	S4 internal-state dynamics / collapse diagnostics
timeseries_Prisoners_Dilemma_hybrid-neutral-neutral_vs_fixed-tit_for_tat_agent1.png	results/02_paper/20260511_211733_paper_S4	S4 internal-state dynamics / collapse diagnostics
cumulative_payoff_Prisoners_Dilemma_hybrid-neutral-neutral_vs_rational.png	results/02_paper/20260511_211733_paper_S4	S4 internal-state dynamics / collapse diagnostics
event_values_Prisoners_Dilemma_hybrid-neutral-neutral_vs_rational_agent1.png	results/02_paper/20260511_211733_paper_S4	S4 internal-state dynamics / collapse diagnostics

9. Рекомендуемая формулировка общего вывода

Paper runs substantially strengthen the empirical section by replacing isolated examples with reproducible multi-seed runs over cross-type matchups, fixed-strategy adaptation, personality/intensity sweeps, and internal-state dynamics. The results support a descriptive claim that payoff-only evaluation is insufficient: in several conditions high external payoff coexists with degraded internal state, asymmetric exploitation, or poor game-specific social metrics. However, the current experiments should not be presented as proof of human-like behavior or as validation of future white-box/belief-update components.

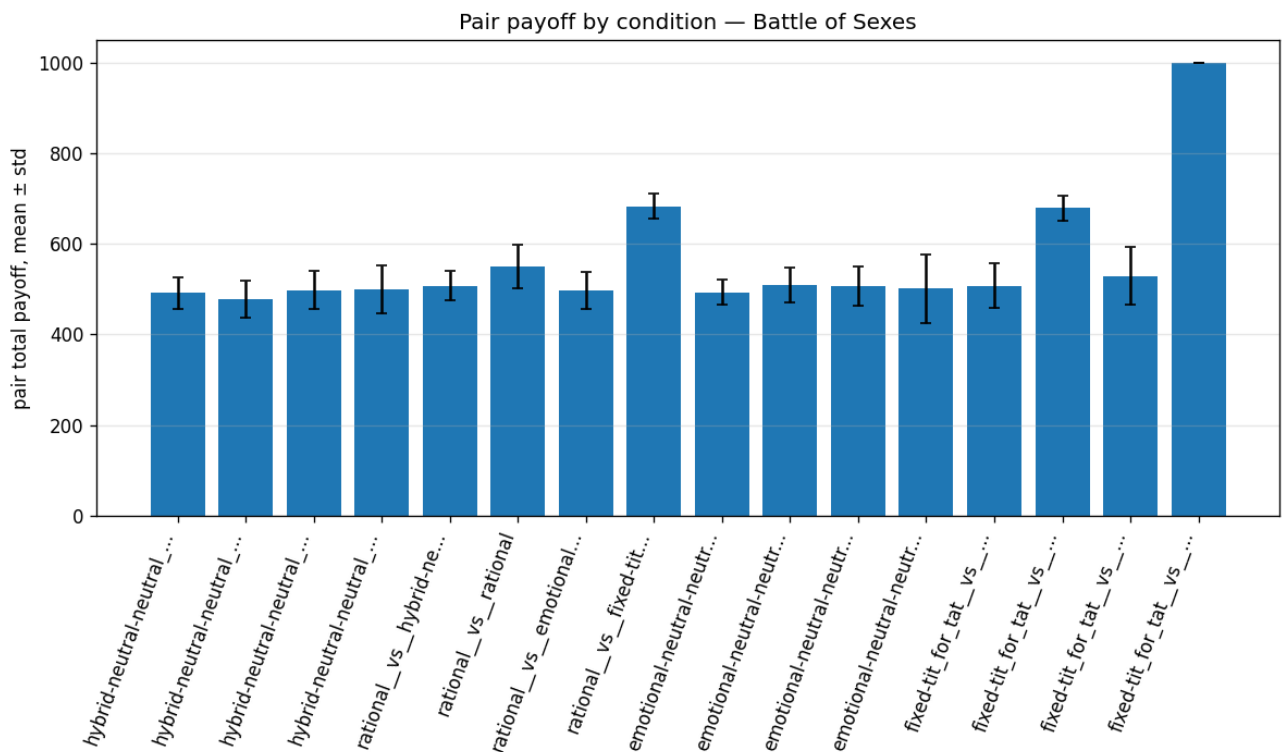
Appendix: selected figures

payoff_bar_Prisoners_Dilemma.png



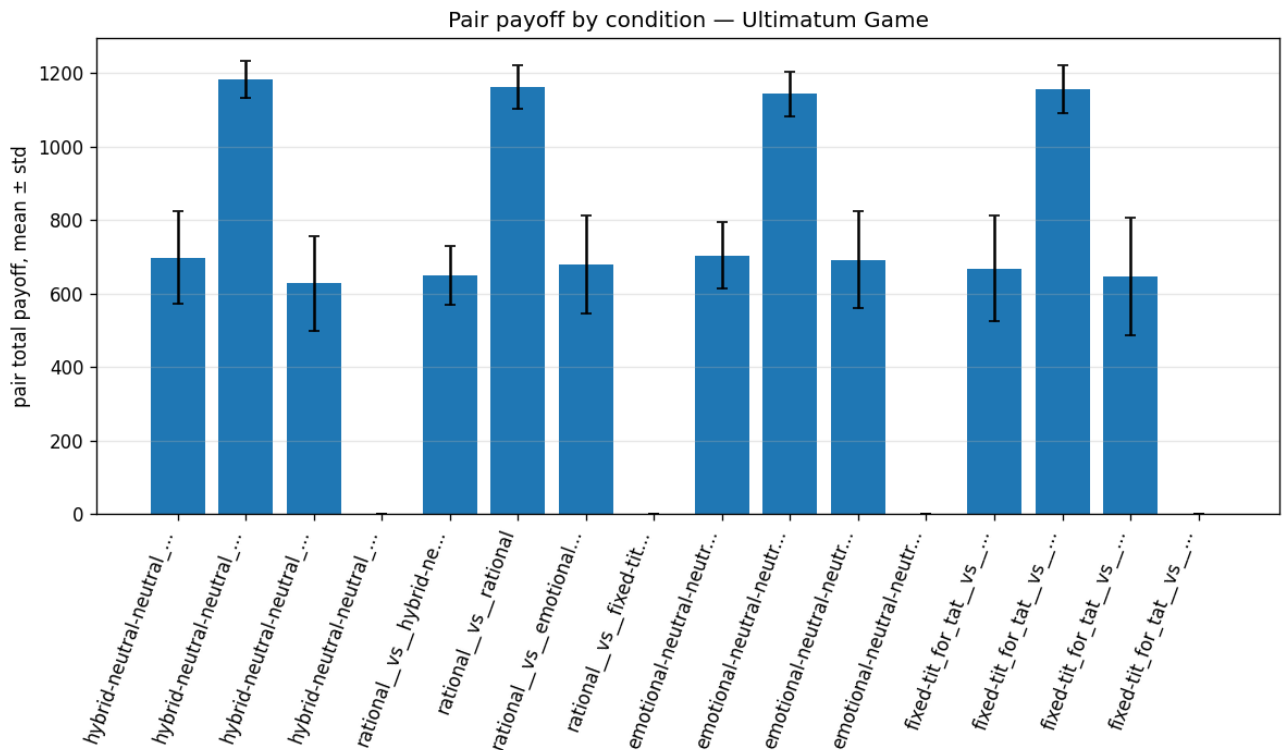
Purpose: S1 matchup matrix: payoff/internal state/game-specific metric

payoff_bar_Battle_of_Sexes.png



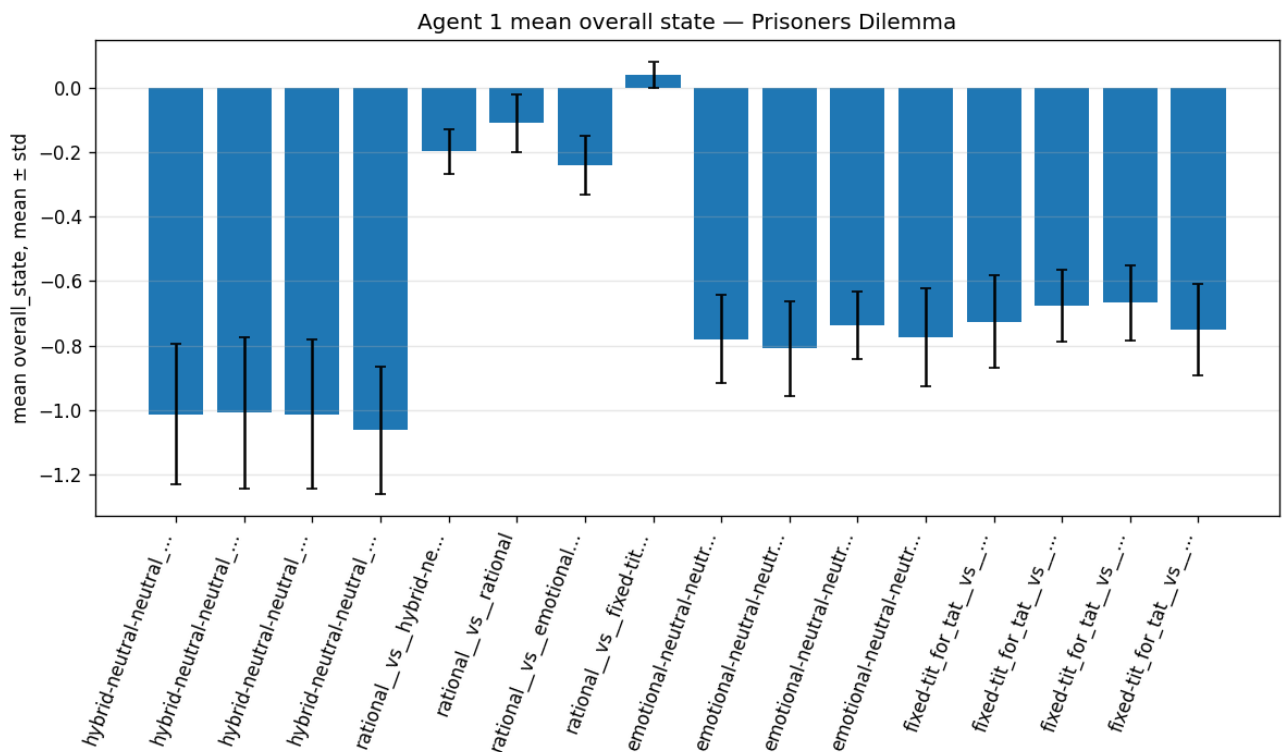
Purpose: S1 matchup matrix: payoff/internal state/game-specific metric

payoff_bar_Ultimatum_Game.png



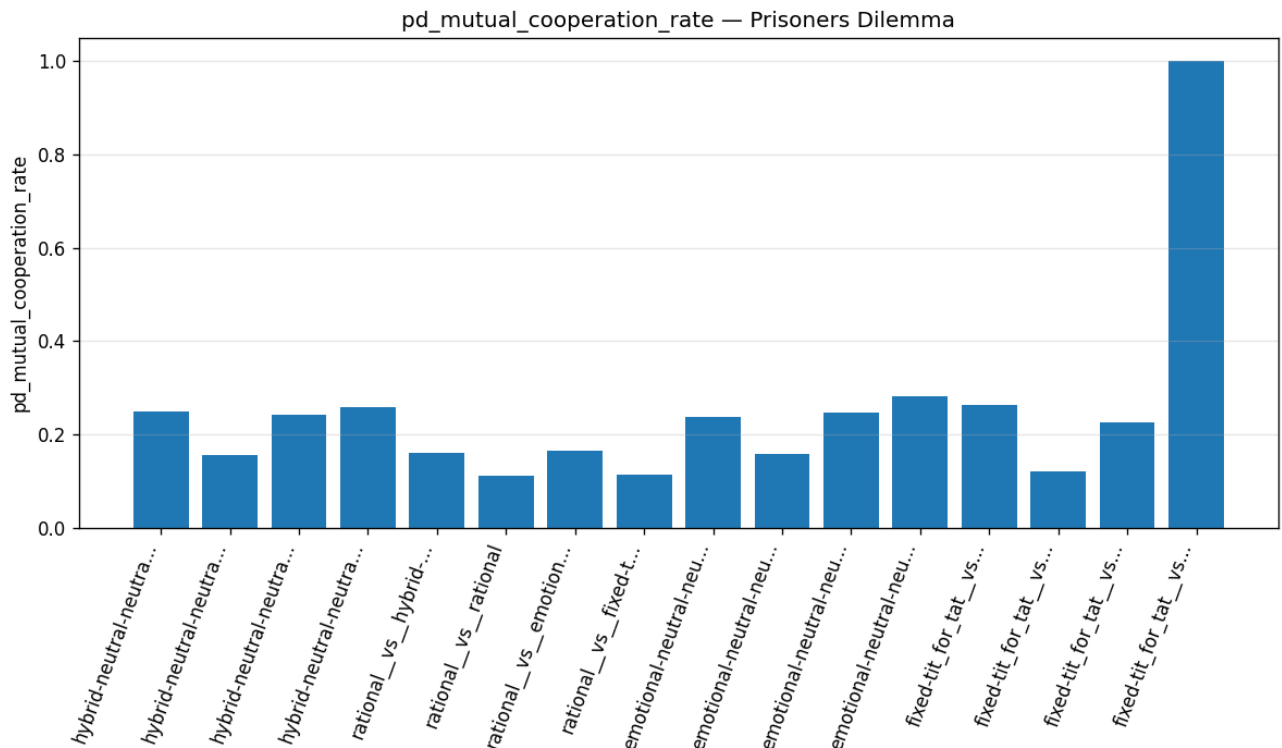
Purpose: S1 matchup matrix: payoff/internal state/game-specific metric

overall_state_bar_Prisoners_Dilemma.png



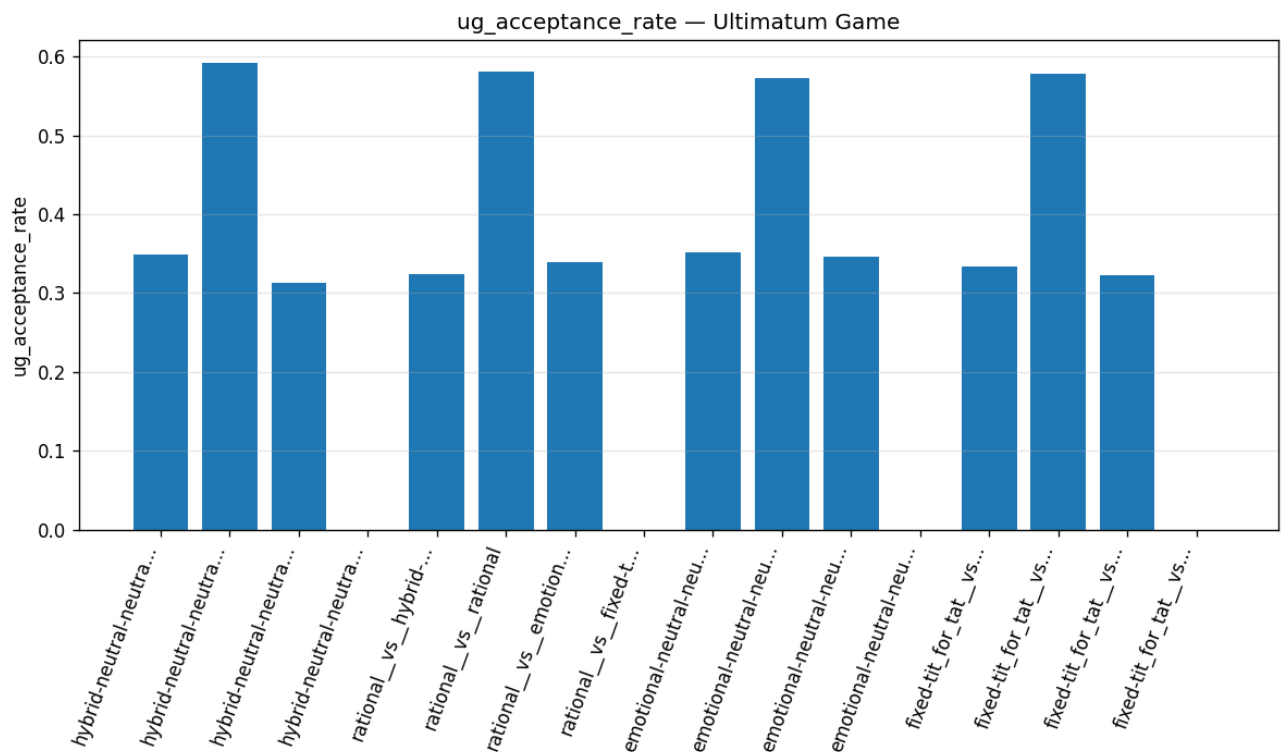
Purpose: S1 matchup matrix: payoff/internal state/game-specific metric

game_metric_Prisoners_Dilemma_pd_mutual_cooperation_rate.png



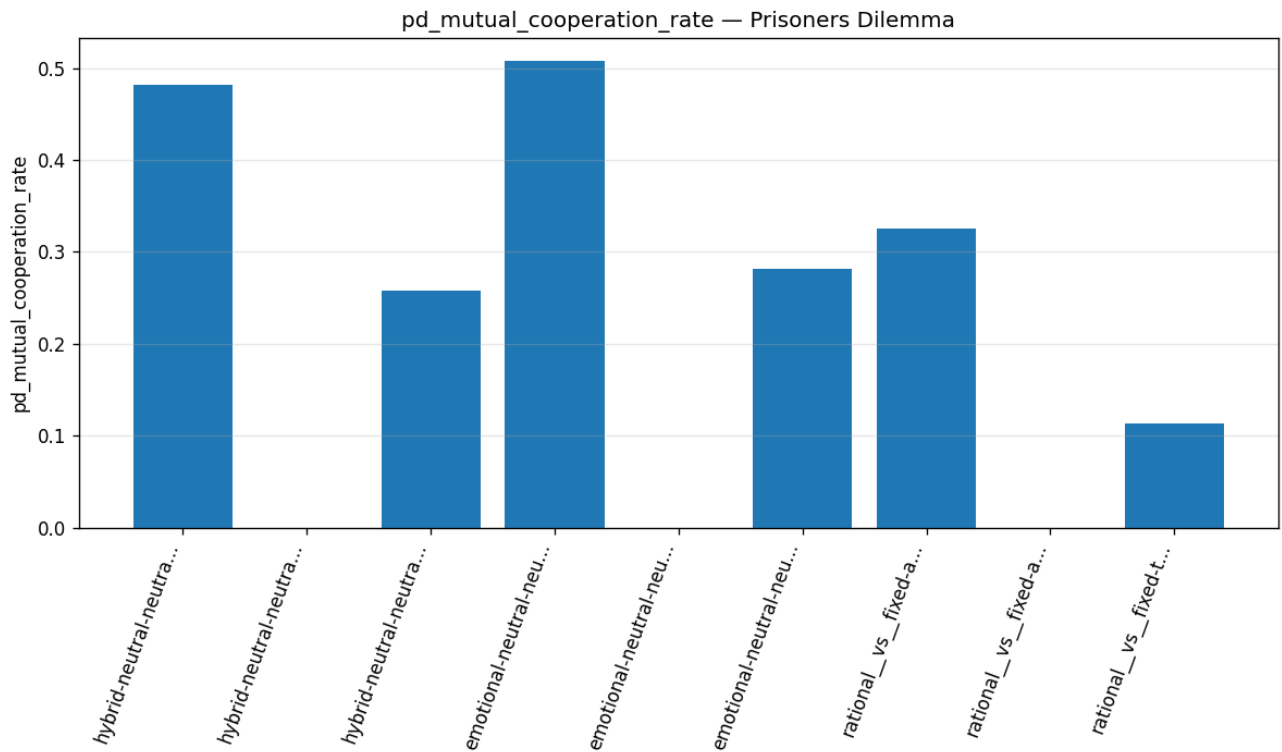
Purpose: S1 matchup matrix: payoff/internal state/game-specific metric

game_metric_Ultimatum_Game_ug_acceptance_rate.png



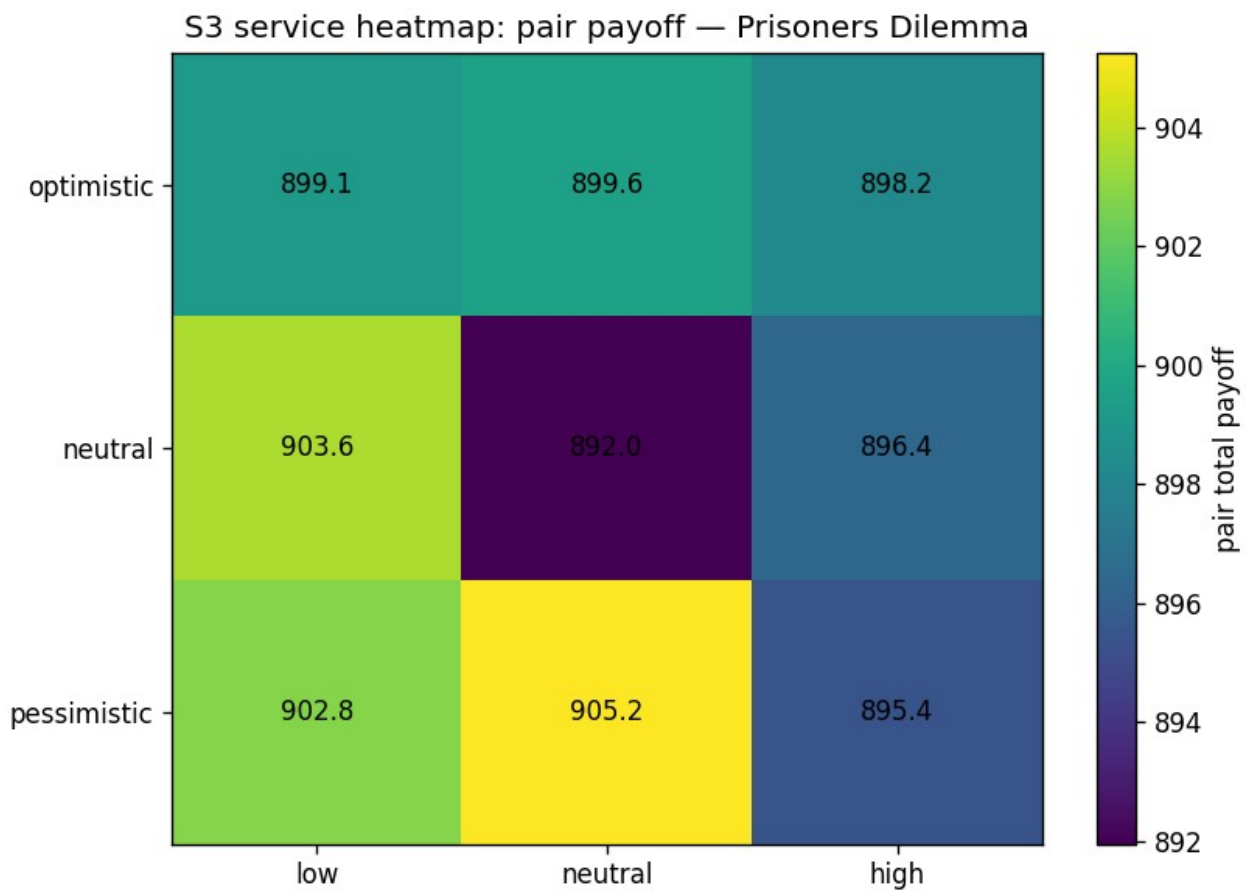
Purpose: S1 matchup matrix: payoff/internal state/game-specific metric

adaptation_first_vs_last.png



Purpose: S2 adaptation to fixed strategies

S3_heatmap_pair_payoff_Prisoners_Dilemma.png



Purpose: S3 personality x intensity core PD